

# RBF-FD Solver: Least Squares Method for Operator Reproduction

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9/2/2026

## 1 Problem Setting:

Following the standard RBF-FD methods, let  $u \in \mathcal{F}$  be a solution for a PDE in form of:

$$-Lu = -\nabla \cdot (A \nabla u) = f \quad \text{in } \Omega, \quad (1)$$

Let  $1 < p < \infty$ ,  $A \in C^{0,1}(\bar{\Omega}; \mathbb{R}^{d \times d}) : \bar{\Omega} \rightarrow \mathbb{R}^{d \times d}$  is an symmetric positive definite anisotropic tensor, where

$$\eta |\xi|^2 \leq \mathbf{A}(\mathbf{x}) \xi \cdot \xi \leq \mathbf{H} |\xi|^2, \quad \mathbf{0} < \eta \leq \mathbf{H} \quad (2)$$

and  $f \in L^p(\Omega)$  is the forcing term acting in the domain  $\Omega$ . The problem is subject to Neumann and Dirichlet Boundary Conditions partitioned within  $\partial\Omega$  of the form:

$$u = g_D \quad \text{on } \Gamma_D \subset \partial\Omega, \quad (3)$$

$$\frac{\partial u}{\partial n} \equiv n \cdot A \nabla u = g_N \quad \text{on } \Gamma_N \subset \partial\Omega, \quad (4)$$

For simplicity, we assume  $\partial\Omega = \Gamma_D \cup \Gamma_N$  and  $\Gamma_D \cap \Gamma_N = \emptyset$ . Furthermore, let  $g_D, g_N \in W^{k-1,p}(\partial\Omega)$ . By *Agmon-Douglis-Nirenberg* Theorem [2], we have the following estimate for  $u$ :

$$\|u\|_{W^{k,p}(\Omega)} \leq C \left( \|Lu\|_{L^p(\Omega)} + \sum_{j=1}^m \|g_j\|_{W^{k-1,p}(\partial\Omega)} + \|u\|_{L^p(\Omega)} \right). \quad (5)$$

## 2 RBF-FD Method:

### 2.1 Local Approximation setup

To obtain an approximation of the operator  $L$  at point  $\mathbf{z}$ , we can approximate  $\hat{u} \approx u$  on a neighborhood  $\Omega \subseteq \mathbb{R}^d$  of  $\mathbf{z}$  such that we approximate  $Lu \approx L\hat{u}$ . We define the radial basis functions in the form  $\phi_i(\mathbf{x}) := \phi(\|\mathbf{x} - \mathbf{y}_i\|) : \Omega \rightarrow \mathbb{R}$  for  $i = 1, \dots, m$  and centers  $\{\mathbf{y}_i\}_{i=1}^m \subseteq \Omega$ . Define the vector space  $V_Y = \text{span}\{\phi_1, \dots, \phi_m\}$ . Let  $X = \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subseteq \Omega$  be a set of stencil nodes for a point  $\mathbf{z}$  in  $\Omega$ . For the classical approximation scheme, let  $X = Y$  and  $m = n$ . We can now define  $\Phi \in \mathbb{R}^{n \times m}$ , where  $\Phi_{ij} = \phi_j(\mathbf{x}_i)$ . The role of these basis functions will be to approximate the operator  $L$ . The approximation of  $u$  takes the form of:

$$\hat{u}(\mathbf{x}) = \sum_{i=1}^n \lambda_i \phi_i(\mathbf{x}) \quad (6)$$

This will lead to the form  $\Phi \lambda = \hat{\mathbf{u}}$  for  $\hat{\mathbf{u}} = [u(\mathbf{x}_1), \dots, u(\mathbf{x}_n)]^T$ . We can simply for  $\lambda$  to obtain:

$$\hat{u}(\mathbf{x}) = \sum_{i=1}^n (\Phi^{-1} \mathbf{u})_i \phi_i(\mathbf{x}) = \phi(\mathbf{x})^T \Phi^{-1} \mathbf{u} \quad (7)$$

The approximation of  $Lu$  at  $z$  is in the form of:

$$Lu(\mathbf{z}) \approx L\hat{u}(\mathbf{z}) = \mathbf{L} \phi^T(\mathbf{z}) \lambda = \mathbf{L} \phi^T(\mathbf{z}) \Phi^{-1} \mathbf{u} := \mathbf{w}^{\mathbf{L}, \mathbf{X}}(\mathbf{z}) \mathbf{u} \quad (8)$$

Thus, approximation the operator turns into local solver for the weights  $w_i$  from the form:

$$\Phi^T \mathbf{w}^{\mathbf{L}, \mathbf{X}}(\mathbf{z}) = \mathbf{L} \phi(\mathbf{z}) \quad (9)$$

Through the method of undefined coefficients on the basis  $\{\phi_i\}$  we get the same resulting set of weights. Finally, by defining the basis for  $V_Y$  as radial basis functions results in the Radial Basis Function generated Finite Difference method (RBF-FD).

## 2.2 Polynomial Augmentation

For conditionally positive definite kernels, we augment them through the form:

$$\hat{u}(\mathbf{x}) = \sum_{\mathbf{j}=1}^n \lambda_{\mathbf{i}} \phi_{\mathbf{i}}(\mathbf{x}) + \sum_{\mathbf{j}=1}^l \beta_{\mathbf{j}} \mathbf{p}_{\mathbf{j}}(\mathbf{x}) \quad (10)$$

We can generate a Gram matrix equation of the form:

$$\begin{bmatrix} \hat{\mathbf{u}} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{\Phi} & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{\lambda} \\ \boldsymbol{\alpha} \end{bmatrix} \quad (11)$$

Hence, we can approximate the operator to take the form:

$$(L\hat{u})(\mathbf{x}) = \begin{bmatrix} (L\phi)(\mathbf{x}) \\ (L\mathbf{p})(\mathbf{x}) \end{bmatrix}^T \begin{bmatrix} \mathbf{\Phi} & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix}^{-1} \begin{bmatrix} \hat{\mathbf{u}} \\ \mathbf{0} \end{bmatrix} := \mathbf{w}^T(\mathbf{x}) \hat{\mathbf{u}} \quad (12)$$

The polynomial constraint setup can be viewed as a Lagrange multiplier to a minimization scheme for the weights.

## 2.3 Consistency Error Analysis

We begin by defining the native space for the set of solutions for (1) with (4). Let  $\mathcal{F}_{\mathcal{K}}$  be the RKHS Native space for the set of solutions. Suppose  $V_Y \subseteq V = \text{span}\{\phi(\cdot, \mathbf{x}) : \mathbf{x} \in \Omega \subseteq \mathbb{R}^d\} \subseteq \mathcal{F}_{\mathcal{K}}$ . Let  $f \in \mathcal{F}_{\mathcal{K}}$ , then  $f(\mathbf{x}) = \langle f, \phi(\cdot, \mathbf{x}) \rangle_{\mathcal{F}_{\mathcal{K}}}$ . Then  $(Lf)(\mathbf{x}) = \langle f, L\phi(\cdot, \mathbf{x}) \rangle_{\mathcal{F}_{\mathcal{K}}}$ . Additionally,  $\mathbf{w}^T(\mathbf{x})\mathbf{f} = \langle f, \mathbf{w}^T(\mathbf{x})\phi \rangle_{\mathcal{F}_{\mathcal{K}}}$ . Defining the error functional  $\xi_{w,x}(f) = \sup_{f \in \mathcal{F}_{\mathcal{K}}, \|f\|_{\mathcal{F}_{\mathcal{K}}} \leq 1} |Lf(\mathbf{x}) - \mathbf{w}^T(\mathbf{x})\mathbf{f}| < \infty$  and  $\xi_{w,x} \in \mathcal{F}_{\mathcal{K}}^*$ . Therefore, by Riesz Representation Theorem,  $g_{w,x} = L\phi(\cdot, \mathbf{x}) - \mathbf{w}^T(\mathbf{x})\phi$ ,  $\xi_{w,x} = \|g_{w,x}\|_{\mathcal{F}_{\mathcal{K}}}$ . Finally, We define the power function  $\mathcal{P}^2(\mathbf{x}) = \|g_{w,x}\|_{\mathcal{F}_{\mathcal{K}}}^2$ .

## 3 Weight Constraint RBF-FD Setup:

Rather than assuming a form for the solution, our goal is to approximate the operator through weights dual to  $u$  sampled at nodes. Suppose  $m > n$  (for the case of  $m < n$ , refer to [1]), thus more center nodes than stencil nodes per stencil. Additionally, instead of assuming exactness of the form  $L\phi_j(\mathbf{x}) = \sum_{i=1}^n w_i(\mathbf{x})\phi_j(\mathbf{x}_i)$  for  $j = 1, \dots, m$  as in the classical RBF-FD, we define a residual function:

$$r_j(\mathbf{x}) := L\phi_j(\mathbf{x}) - \sum_{i=1}^n w_i(\mathbf{x})\phi_j(\mathbf{x}_i) \quad (13)$$

$$r(\mathbf{x}) = L\phi(\mathbf{x}) - \mathbf{\Phi}^T \mathbf{w}(\mathbf{x}), \quad \mathbf{w} : \mathbb{R}^n \rightarrow \mathbb{R}^n \text{ for } \mathbf{x} \in \Omega. \quad (14)$$

Suppose  $\{p_i\}_{i=1}^l$  is a set of polynomials augmenting the radial basis functions. Let  $\Pi_k = \text{span}\{p_i\}_{i=1}^l$  with polynomial degree up to  $k$ . We want to reproduce the operator exactly on the polynomial basis:

$$Lp_j(\mathbf{x}) := \sum_{i=1}^n w_i(\mathbf{x})p_j(\mathbf{x}_i) \text{ for } j = 1, \dots, l \quad (15)$$

$$L\mathbf{p}(\mathbf{x}) = \mathbf{P}^T \mathbf{w} \quad (16)$$

where  $\mathbf{P}_{ij} := p_j(\mathbf{x}_i) \in \mathbb{R}^{n \times l}$ . Thus, we have a constrained least squares problem in the form:

$$\begin{cases} \min J(\mathbf{w}(\mathbf{x})) := \min \frac{1}{2} \|\mathbf{r}(\mathbf{x})\|_2^2 & \text{for } \mathbf{x} \in \Omega \\ \text{subject to } \Phi^T \mathbf{w} = L\mathbf{p} \end{cases} \quad (17)$$

We can solve this setup by defining the Lagrangian  $\mathcal{L}$  for the problem as follows:

$$\begin{aligned} \mathcal{L}(\mathbf{w}, \boldsymbol{\nu}) &:= J(\mathbf{w}) + \langle \boldsymbol{\nu}, \mathbf{P}^T \mathbf{w} - L\mathbf{p}(\mathbf{x}) \rangle \\ &= \frac{1}{2} \langle \Phi^T \mathbf{w}, \Phi^T \mathbf{w} \rangle - \langle \Phi^T \mathbf{w}, L\phi \rangle + \frac{1}{2} \langle L\phi, L\phi \rangle \\ &\quad + \langle \boldsymbol{\nu}, \mathbf{P}^T \mathbf{w} - L\mathbf{p}(\mathbf{x}) \rangle. \end{aligned}$$

## 4 Consistency Conditions for Finite Spaces and Interpolation Results

The resulting KKT equations for this system are then:

$$\nabla_{\mathbf{w}} \mathcal{L} = 0 : \Phi L\phi = \Phi \Phi^T \mathbf{w} + \mathbf{P}\boldsymbol{\nu} \quad (18)$$

$$\nabla_{\boldsymbol{\nu}} \mathcal{L} = 0 : \mathbf{P}^T \mathbf{w} = L\mathbf{p} \quad (19)$$

The augmented matrix then takes the form:

$$\begin{bmatrix} \Phi \Phi^T & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{w} \\ \boldsymbol{\nu} \end{bmatrix} = \begin{bmatrix} \Phi L\phi \\ L\mathbf{p} \end{bmatrix} \quad (20)$$

From [6], Equation (20) has a unique solution given  $L\mathbf{p} \in \text{Range}(\mathbf{P}^T)$ . The primary goal is to approximate  $Lu(\mathbf{x}) \approx \mathbf{w}^T(\mathbf{x})\mathbf{u}$ , with  $\mathbf{u} = [u(\mathbf{x}_1), \dots, u(\mathbf{x}_n)]^T$ . Define  $L_h f(\mathbf{x}) := \mathbf{w}^T(\mathbf{x})\mathbf{f}$  for some function  $f$ . Thus, we have the following:

$$\begin{aligned} L_h f(\mathbf{x}) &= \mathbf{w}^T(\mathbf{x})\mathbf{f} \\ &= [\mathbf{w}^T \quad \boldsymbol{\nu}^T] \begin{bmatrix} \mathbf{f} \\ \mathbf{0} \end{bmatrix} \\ &= \left( \begin{bmatrix} \Phi \Phi^T & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix}^{-1} \begin{bmatrix} \Phi L\phi \\ L\mathbf{p} \end{bmatrix} \right)^T \begin{bmatrix} \mathbf{f} \\ \mathbf{0} \end{bmatrix} \\ &= [L\phi^T \Phi^T \quad L\mathbf{p}^T] \begin{bmatrix} \Phi \Phi^T & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix}^{-T} \begin{bmatrix} \mathbf{f} \\ \mathbf{0} \end{bmatrix} \\ &= [L\phi^T \Phi^T \quad L\mathbf{p}^T] \begin{bmatrix} \mathbf{q} \\ \boldsymbol{\alpha} \end{bmatrix} \\ &= L\phi^T(\mathbf{x}) (\Phi^T \mathbf{q}) + L\mathbf{p}^T(\mathbf{x}) \boldsymbol{\alpha} \end{aligned}$$

Next, we can show that the coefficients  $\mathbf{q}, \boldsymbol{\alpha}$  relate to the coefficients of  $u$  as follows:

$$\begin{bmatrix} \mathbf{f} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \Phi \Phi^T & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \boldsymbol{\alpha} \end{bmatrix} \quad (21)$$

Define  $V_h = \{\rho(x) : \rho(x) = \sum_{i=1}^m \lambda_i \phi_i(x) \text{ such that } \lambda \in \text{Range}(\Phi^T)\} \subseteq V_Y$ . Let  $s \in V_h + \Pi_k$ . Thus,  $s(x) = \phi^T(x)\Phi^T q + p^T(x)\alpha$  where  $q \in \mathbb{R}^n$  and  $\alpha \in \mathbb{R}^l$ . Then:

$$s = s(X) = \Phi\Phi^T q + P\alpha = f(X) = f \quad (22)$$

$$P^T q = 0 \quad (23)$$

Hence, we get interpolation of  $f$  at the node set  $X$ . Furthermore, from the KKT matrix we can show that  $\Phi r = P\nu$  which implies that  $\Phi r \in \text{range}(P) = (\ker(P^T))^\perp$ . Thus, we get the relationship:

$$r^T(\Phi^T z) = 0 \text{ for } P^T z = 0 \quad (24)$$

Finally, we check the consistency error for  $s \in V_h + \Pi_k$  with  $P^T q = 0$ .

$$\begin{aligned} Ls(x) - L_h s(x) &= L(\phi^T(x)\Phi^T q + p^T(x)\alpha) - w^T(x)s \\ &= L\phi^T(x)\Phi^T q + Lp^T(x)\alpha - w^T(x)(\Phi\Phi^T q + P\alpha) \\ &= L\phi^T(x)\Phi^T q + Lp^T(x)\alpha - w^T(x)\Phi\Phi^T q - w^T(x)P\alpha \\ &= (L\phi(x) - \Phi^T w(x))^T \Phi^T q + (Lp^T(x) - P^T w(x))^T \alpha \\ &= r^T \Phi^T q + (Lp(x) - P^T w(x))^T \alpha \end{aligned}$$

$$Ls(x) - L_h s(x) = 0 \text{ due to (19) and (24)} \quad (25)$$

## 5 Consistency Results for Function Approximations

Let  $u$  be a solution to (1) and  $s \in V_h + \Pi_k$ . We can say  $u = s + (u - s)$  to get the following:

$$\begin{aligned} \mathcal{E}[u](x) &:= Lu(x) - L_h u(x) \\ &= Lu(x) - w^T(x)u \\ &= L((u - s) + s)(x) - w^T(x)u - L_h s(x) + L_h s(x) \\ &= L(u - s)(x) + (Ls(x) - L_h s(x)) + L_h(s - u)(x) \\ &= L(u - s)(x) - w^T(x)(u - s), \text{ by (25)} \\ &= Le(x) - w^T(x)e, e(x) = u(x) - s(x) \\ &= \mathcal{E}[e](x) \end{aligned}$$

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